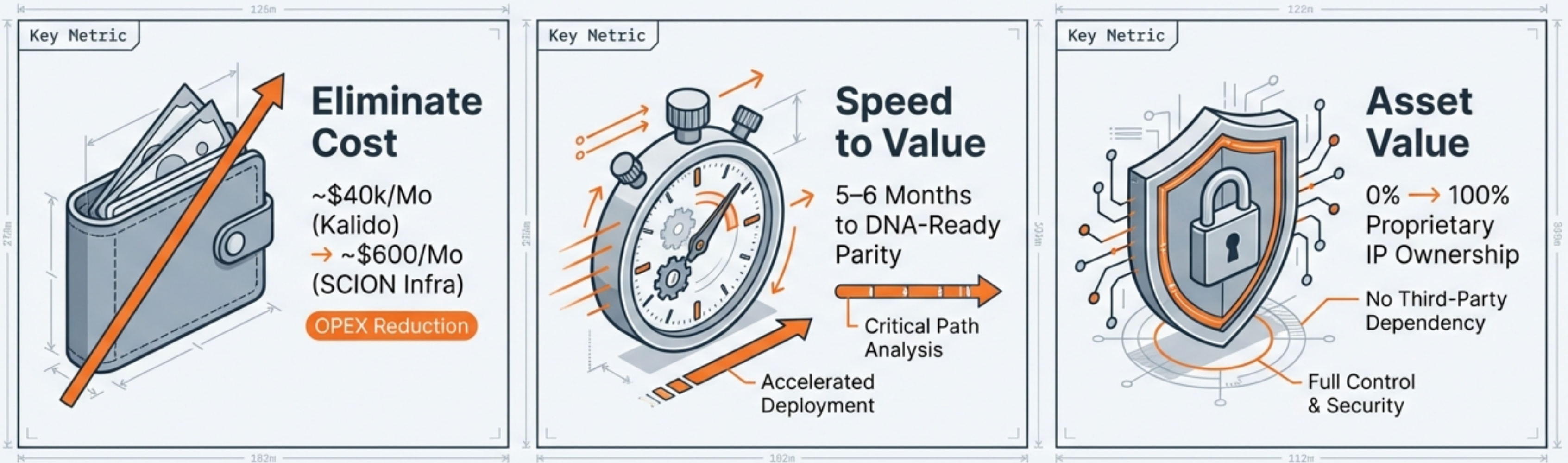


SCION: Scalable Intelligent Governance for Teradata DNA



Transitioning from Legacy Metadata Management
to Proprietary Structural Intelligence.

The Case for Proprietary Governance



EXECUTIVE TAKEAWAYS



The Mandate: Replace opaque 'black-box' legacy tools to eliminate exposure and recurring fees.



The Solution: Deploy SCION—a deterministic, modular governance platform with native AI augmentation.

Engine 1: Metadata Ingestion Layer

Role / Purpose:

The Universal Adapter. Responsible for extracting and normalizing structural metadata from EDW engines.

Core Capabilities:

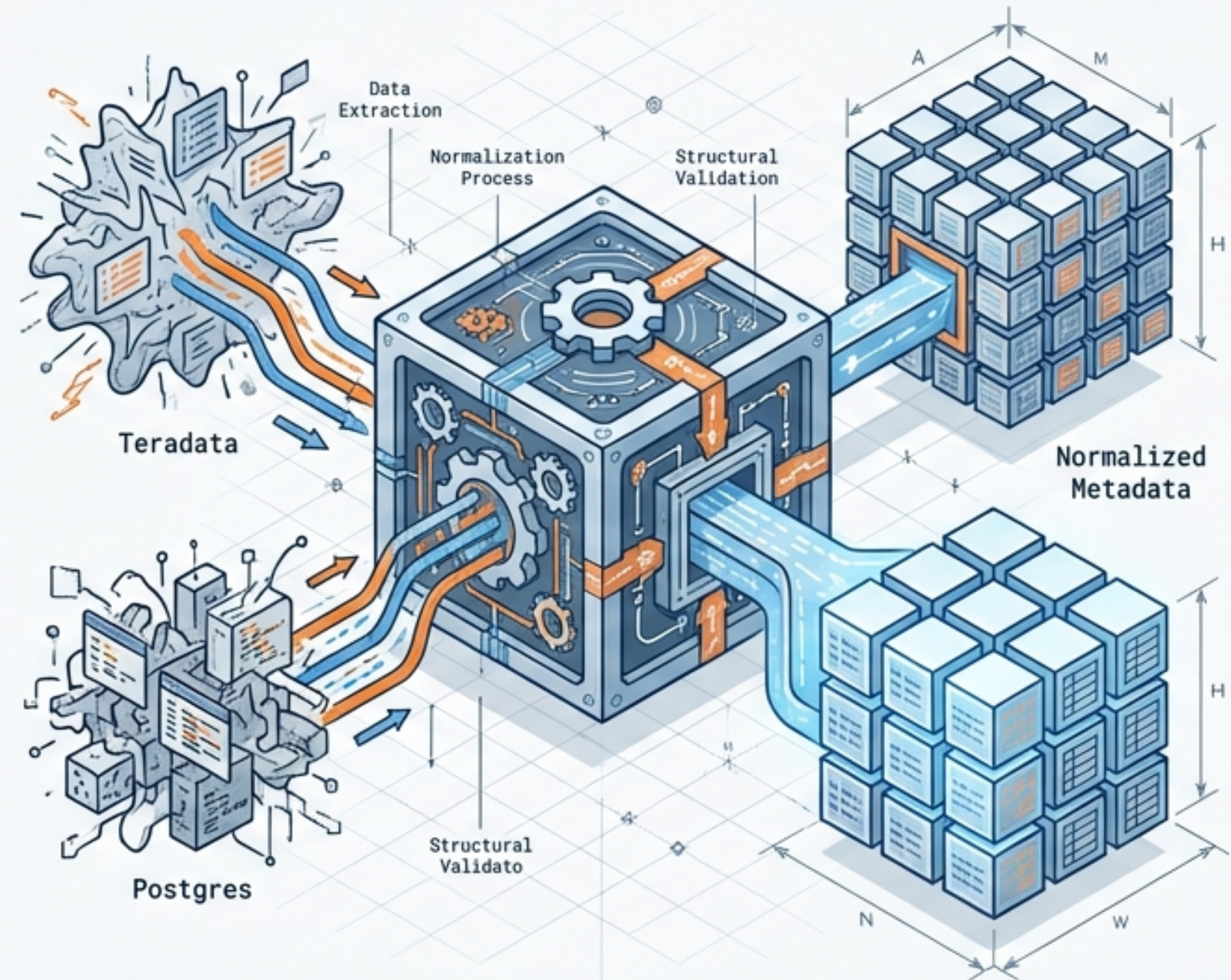
- Source-agnostic extraction (**Teradata/Postgres** adapters).
- Canonical **YAML-based template engine**.
- **Structural validation** and **typological normalization**.

Technical Value:

Decouples governance from specific database vendors; eliminates raw catalog replication.

Strategic Value:

Future-proofs the platform, allowing DNA to adapt to new EDW technologies without re-platforming.



Engine 2: Snapshot Engine

Role / Purpose:

The Time Machine. Captures full EDW state to guarantee transactional consistency.

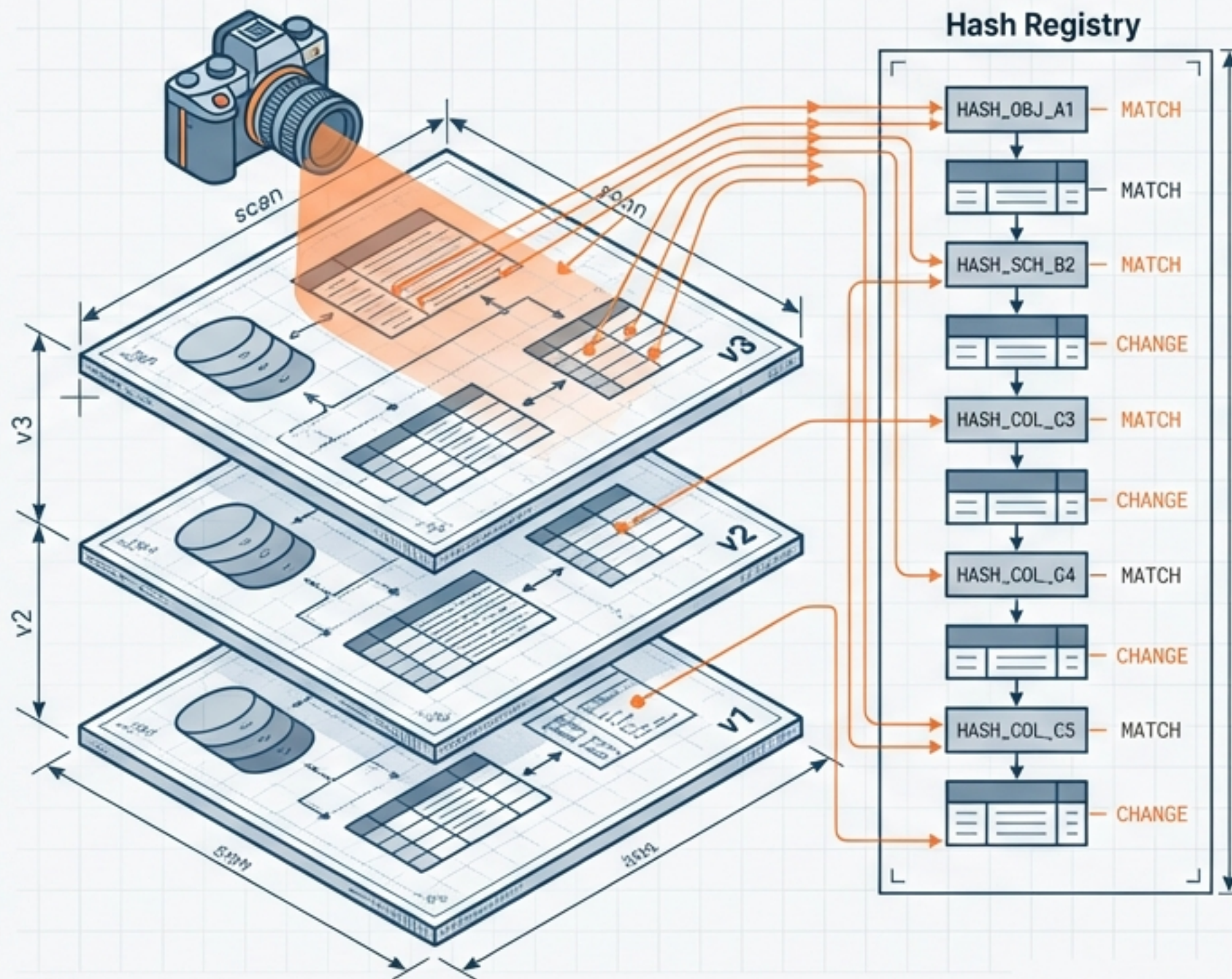
Core Capabilities:

- Historical versioning with structural hashing.
- Detects changes via hash comparison (Zero data read).
- Metrics capture: Object, Schema, and Column counts.

Technical Value: Moves from passive tracking to active state monitoring.

Strategic Value:

Provides the foundational 'truth' required for regulatory compliance and auditability.



Engine 3: Diff Engine

Role / Purpose:

The Change Detective. Compares snapshots to identify and classify structural evolution.

Core Capabilities:

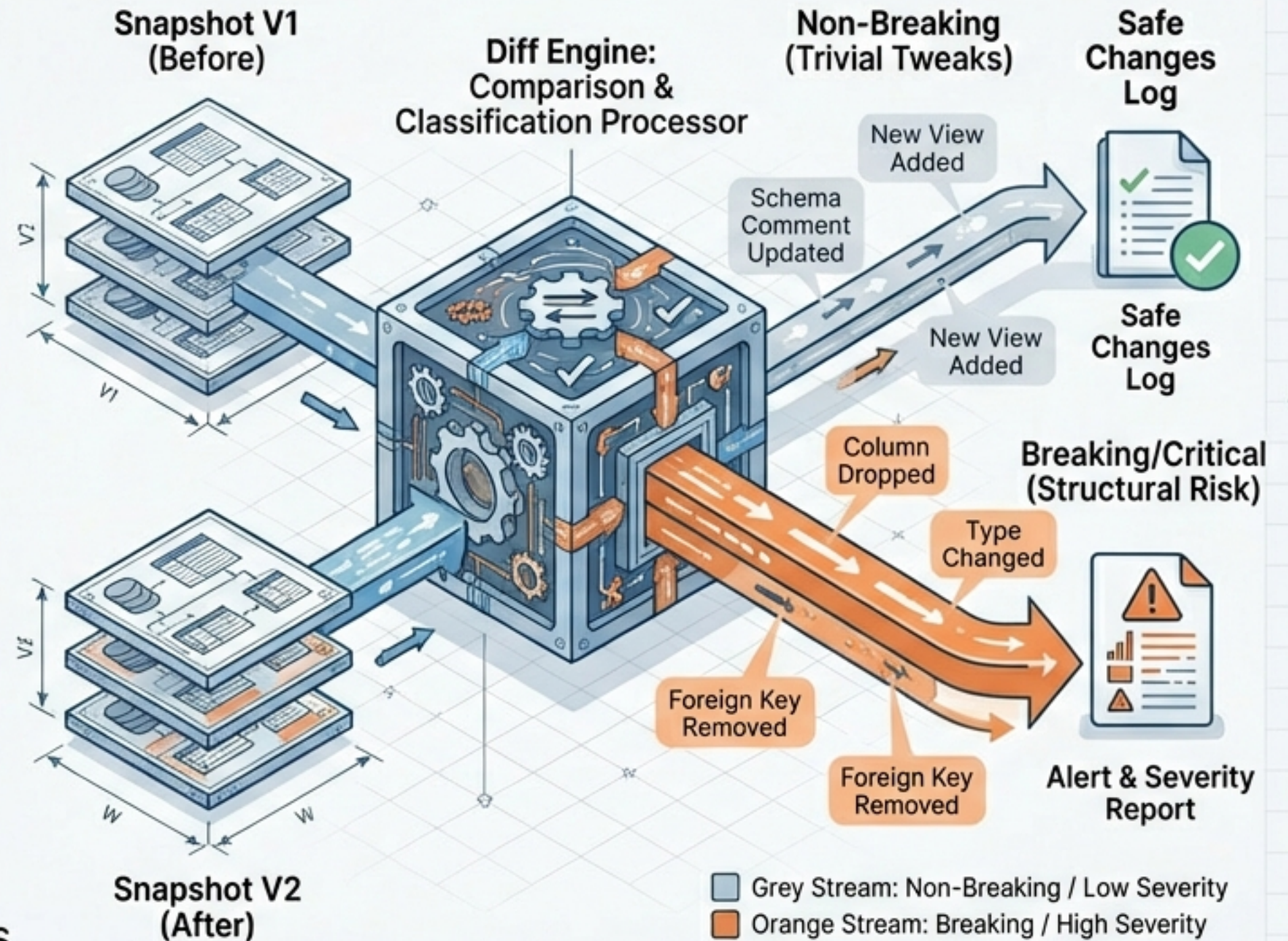
- Deterministic 'Before/After' comparison.
- 'Breaking' vs. 'Non-Breaking' classification.
- Severity scoring and Change Grouping by domain.

Technical Value:

Idempotent processing; produces audit-ready JSON logs of exactly *what* changed. (Using JetBrains Mono for 'JSON')

Strategic Value:

Eliminates 'alert fatigue' by distinguishing between trivial tweaks and critical structural risks.



Engine 4: Graph & Impact Engine

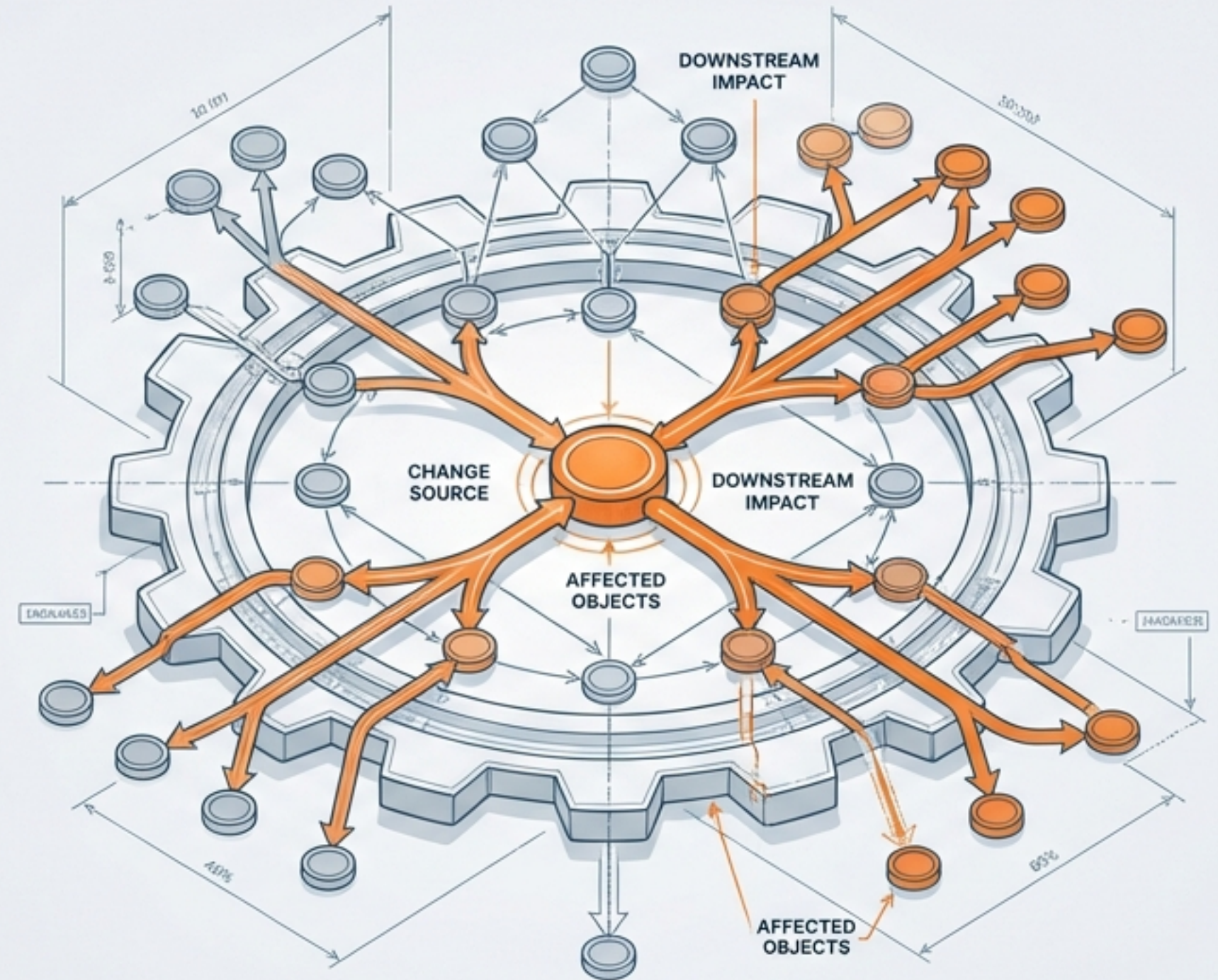
Role / Purpose: The Network Mapper. Traces lineage and calculates the “blast radius” of changes.

Core Capabilities:

- Recursive SQL-based lineage tracing (Nodes & Edges).
- Fragility scoring and Centrality metrics.
- Cycle detection and Critical Path analysis.

Technical Value: Native SQL graph logic removes dependency on external graph databases.

Strategic Value: Proactively predicts outages by visualizing downstream impact before deployment.

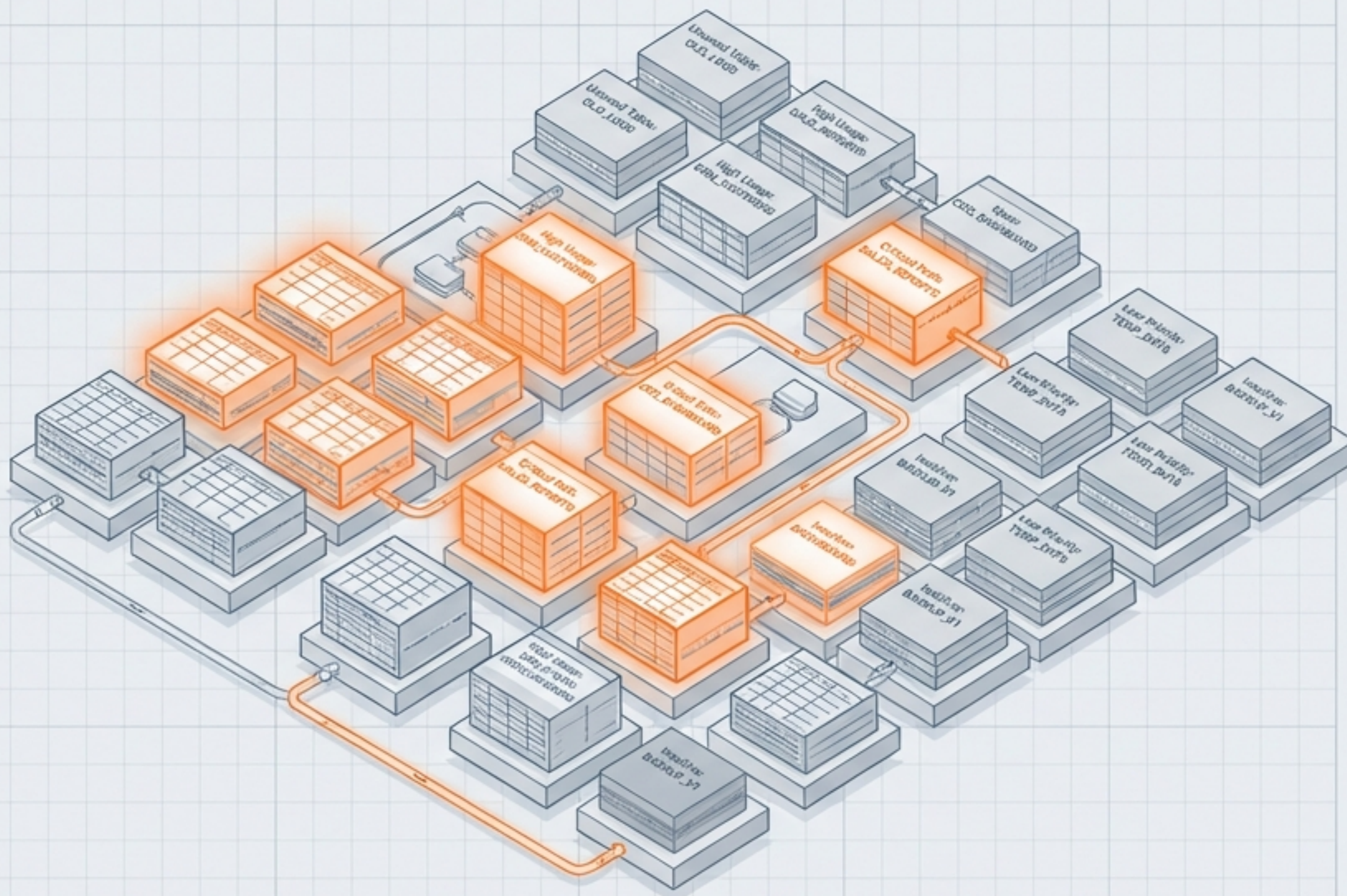


Core Capabilities:

- Query log ingestion and parsing.
- Usage frequency heat-mapping.
- User-weighting (e.g., CEO Report > Ad-hoc query).

Technical Value: Merges static metadata (structure) with dynamic logs (runtime).

Strategic Value: Shifts governance from theoretical impact to actual business risk.



Engine 6: Intelligence Metrics Layer

Role / Purpose:

The Analyst. Aggregates raw signals into high-level ecosystem health indicators.

Core Capabilities:

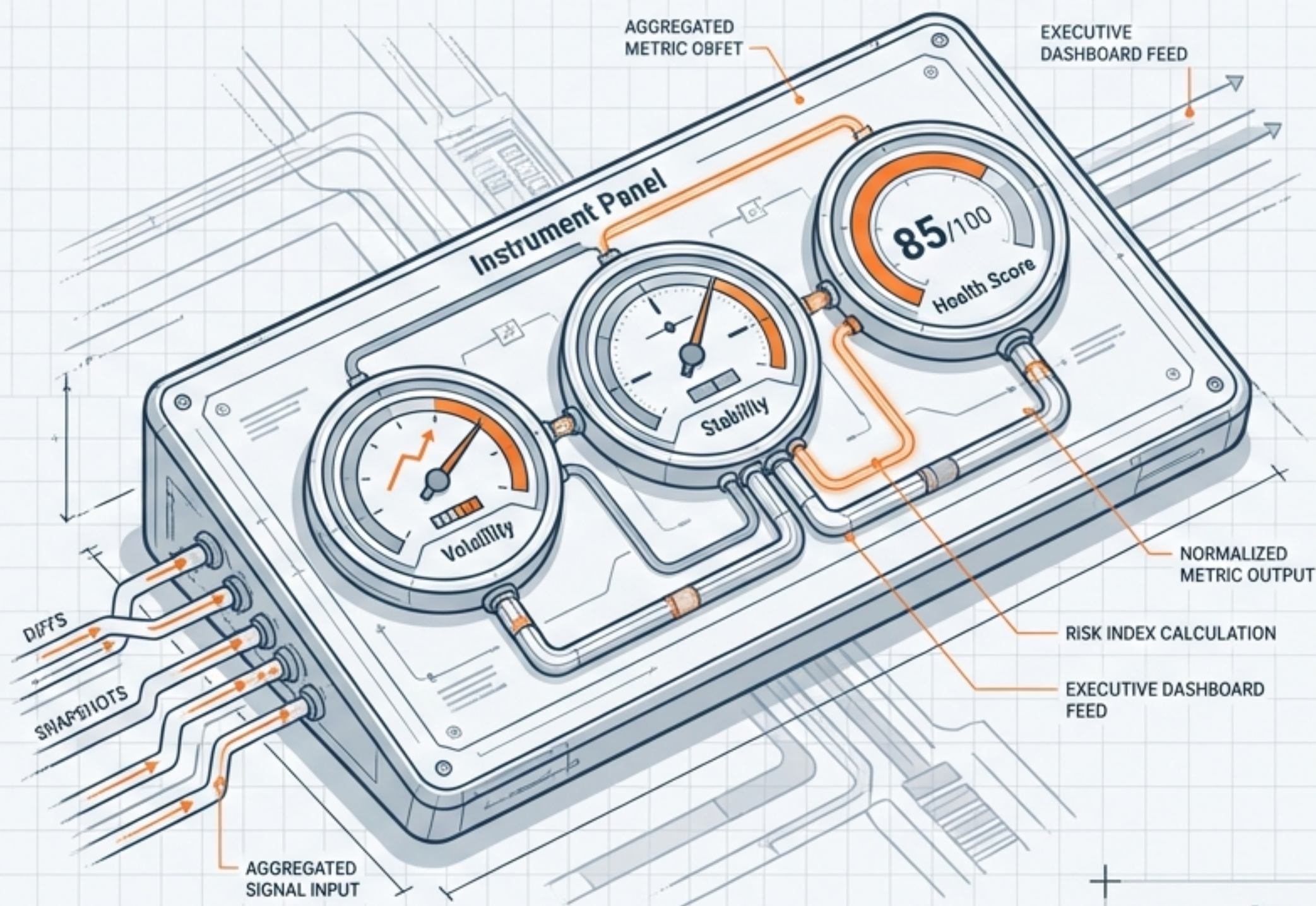
- **Volatility Indexing** (identifying unstable domains). (JetBrains Mono)
- **Stability Trends** per object. (JetBrains Mono)
- **Governance Scorecards** and **Domain Risk Indices**. (JetBrains Mono)

Technical Value:

Transforms disparate data points (diffs, snapshots) into standardized metrics.

Strategic Value:

Elevates SCION from a 'Metadata Tracker' to an 'Enterprise Structural Intelligence Platform'.



Engine 7: TAISA Integration Layer

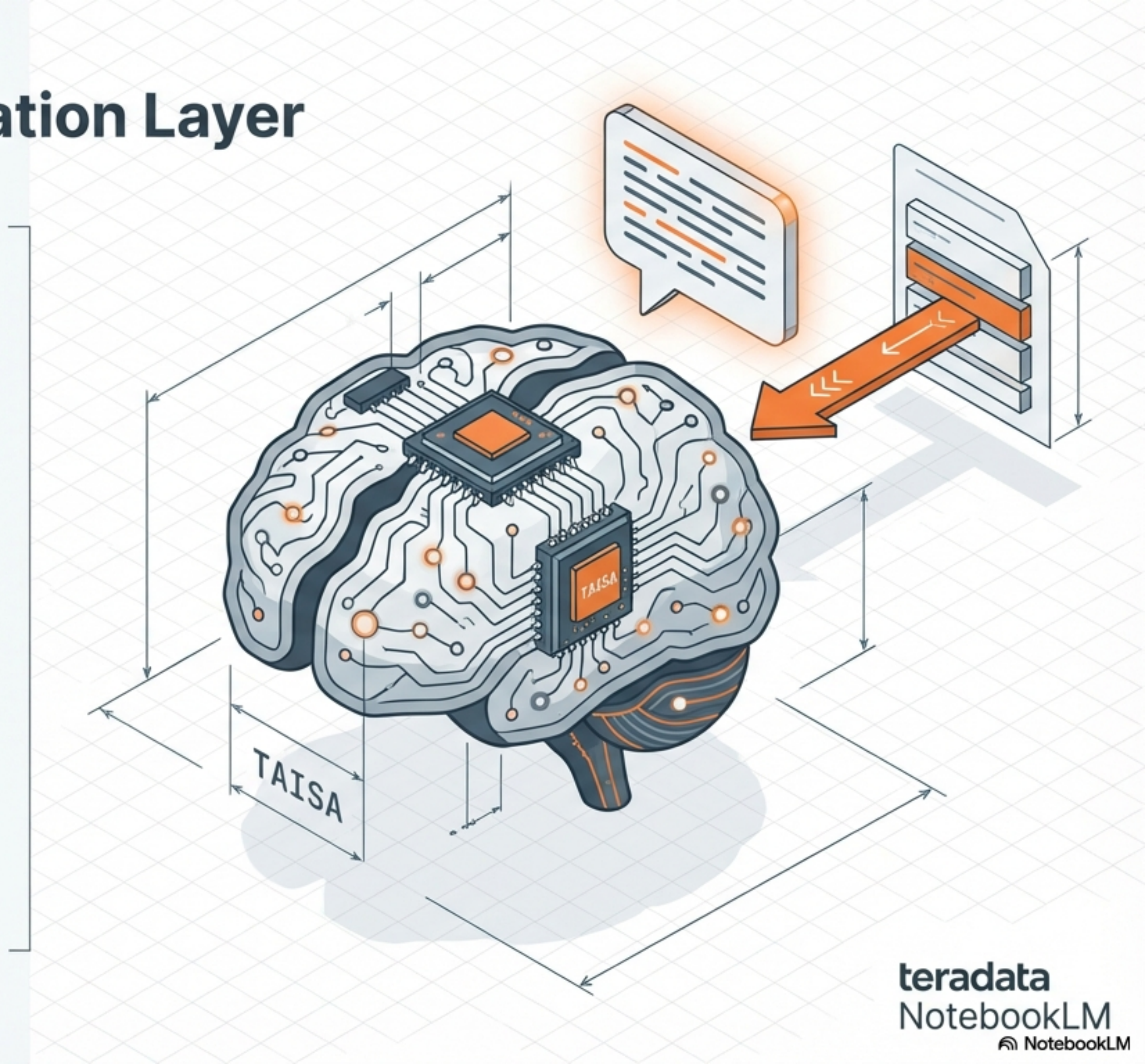
Role / Purpose: The AI Augmentation. Interprets events and explains risk in natural language.

Core Capabilities:

- Stateless API design (Read-Interpret pattern).
- Event interpretation and risk explanation.
- Deterministic context input with versioned reasoning.

Technical Value: AI operates as “Read-Only”—it interprets facts but cannot corrupt state.

Strategic Value: Deploys internal AI (TAISA) as an always-on expert analyst.



Infrastructure & Deployment Requirements

Specs & Runtime

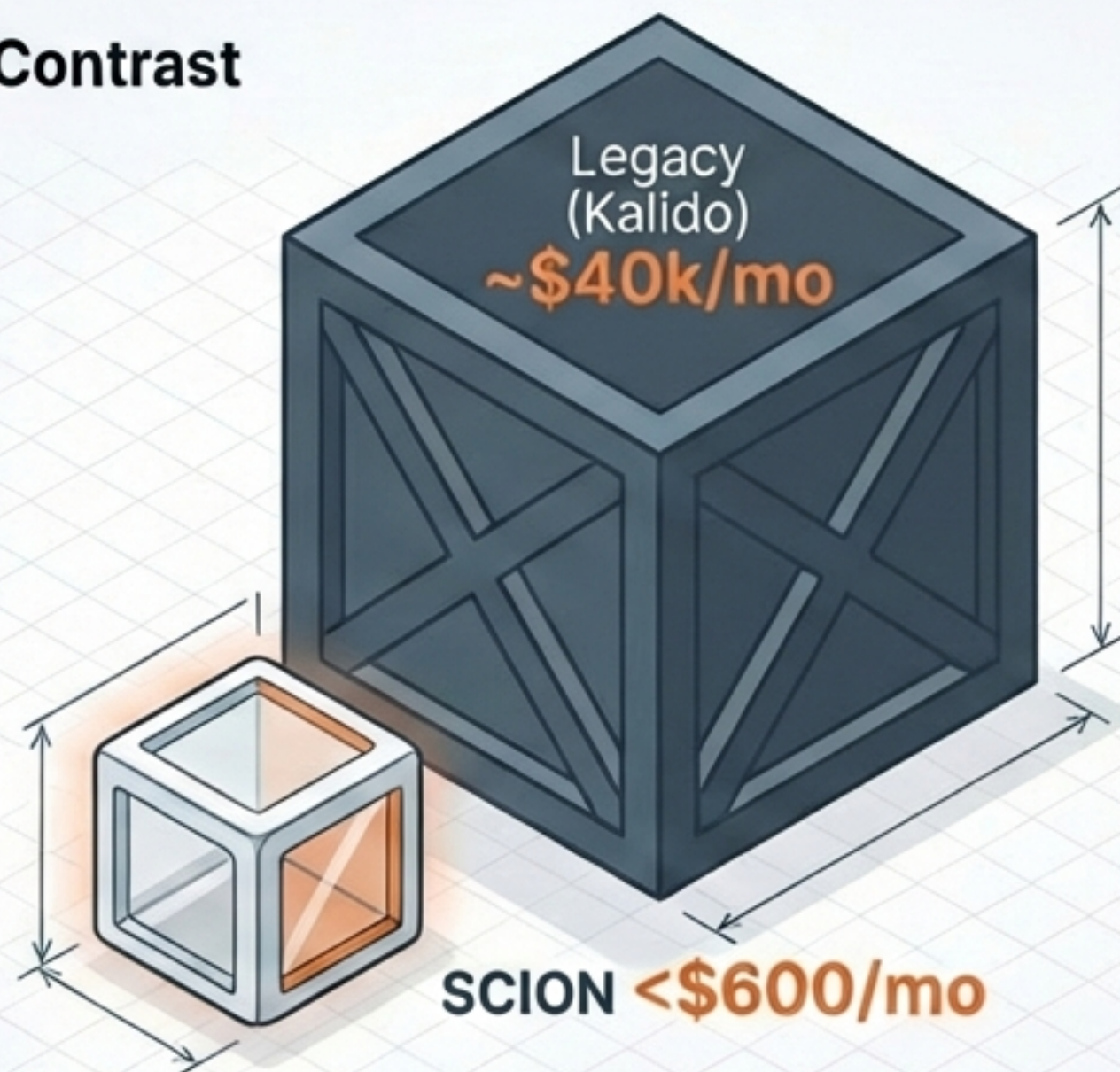
Minimum On-Prem / VM Requirements:

- Compute: 4–8 vCPU
- Memory: 16–32 GB RAM
- Storage: 200–500 GB (Low footprint)
- Runtime: Docker / Fully Containerized

Cloud Instance Estimate:

- ~\$200–\$600/month (Region dependent)
- Scales horizontally via K8s/Docker Swarm

Cost Contrast



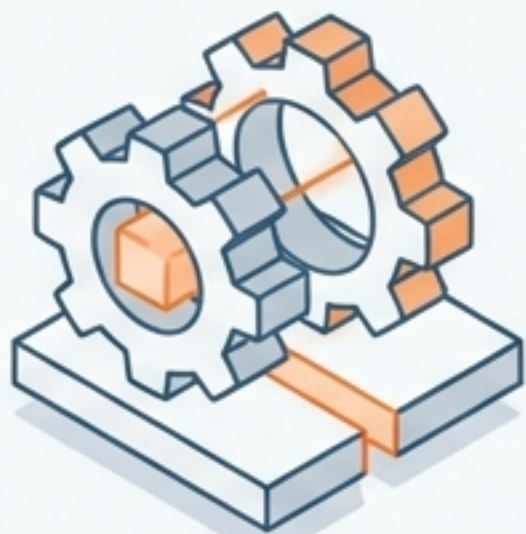
Zero Licensing Fees. No SaaS Dependency.
100% Owned Infrastructure.

teradata

Required Team Profile & Development Model

Internal Engineering Initiative | Not a Consulting Engagement

Lead Engineer (1 FTE)



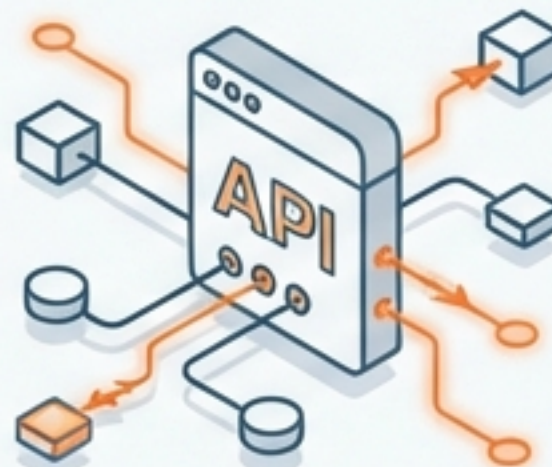
Focus:

- Architecture, **Core Logic** (Diff/Graph), **State Management**.

Responsibility:

System integrity and critical path analysis.

Backend Engineer (1 FTE)



Focus:

- **API Orchestration**, **Ingestion Adapters**, Optimization.

Responsibility:

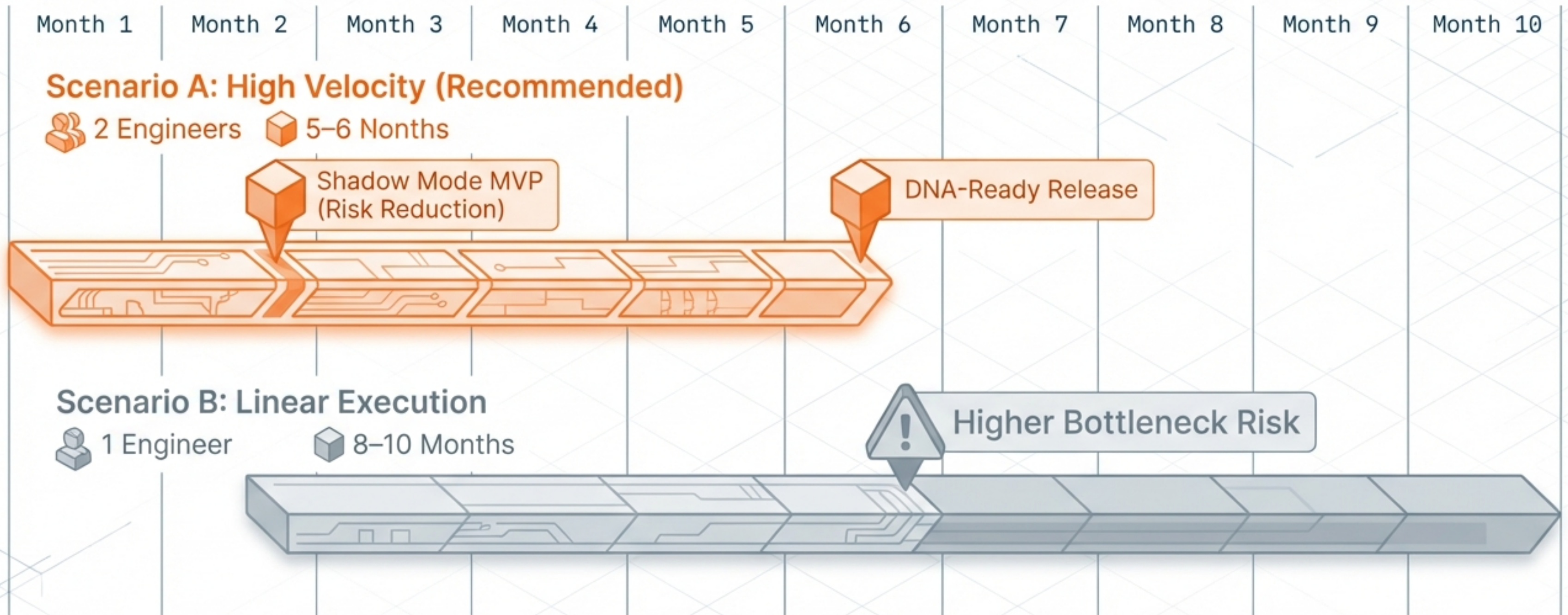
Scalability and data flow.

Internal Continuity:

Development is owned by Teradata DNA. Source code and architectural knowledge are retained internally, eliminating "black-box" dependency.

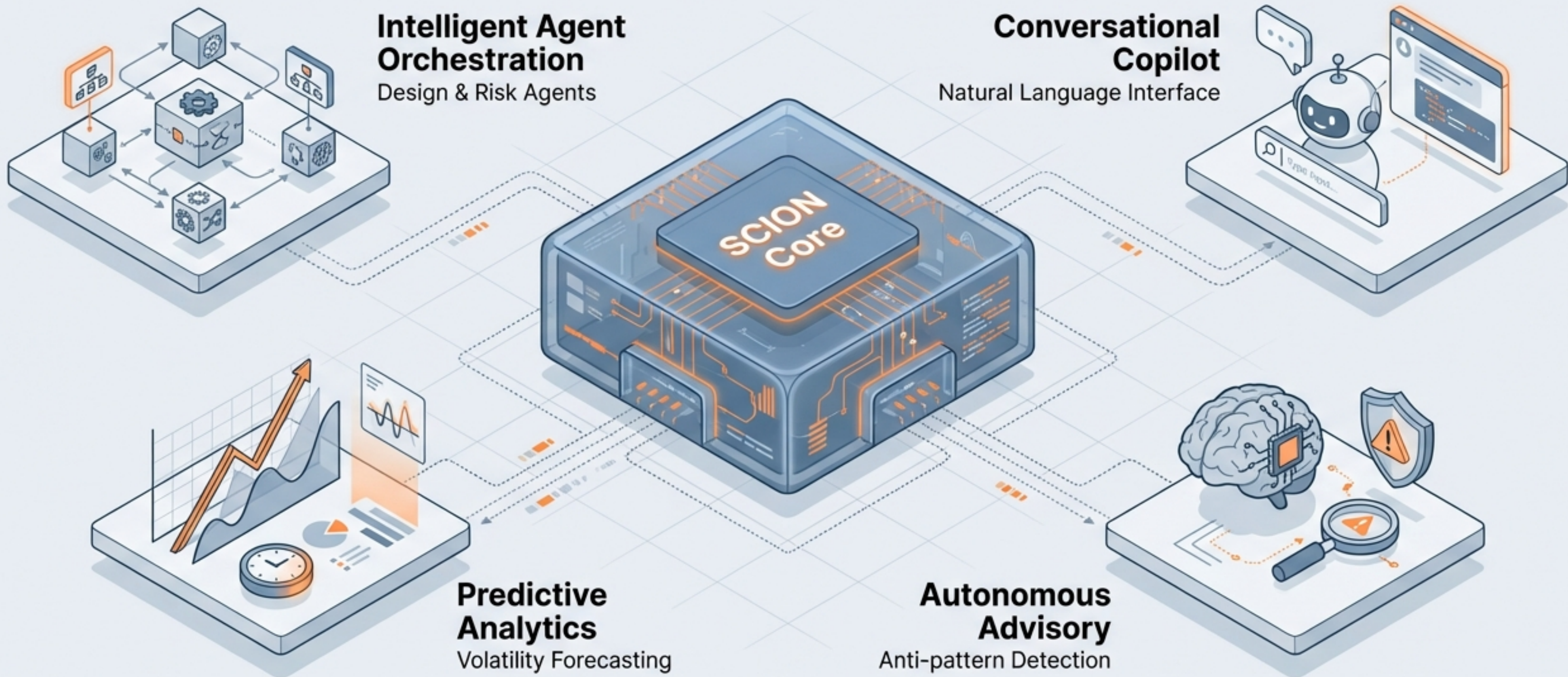


Delivery Timeline & Execution Scenarios



Milestone: MVP Validation at Month 2 ensures architectural viability before full commitment.

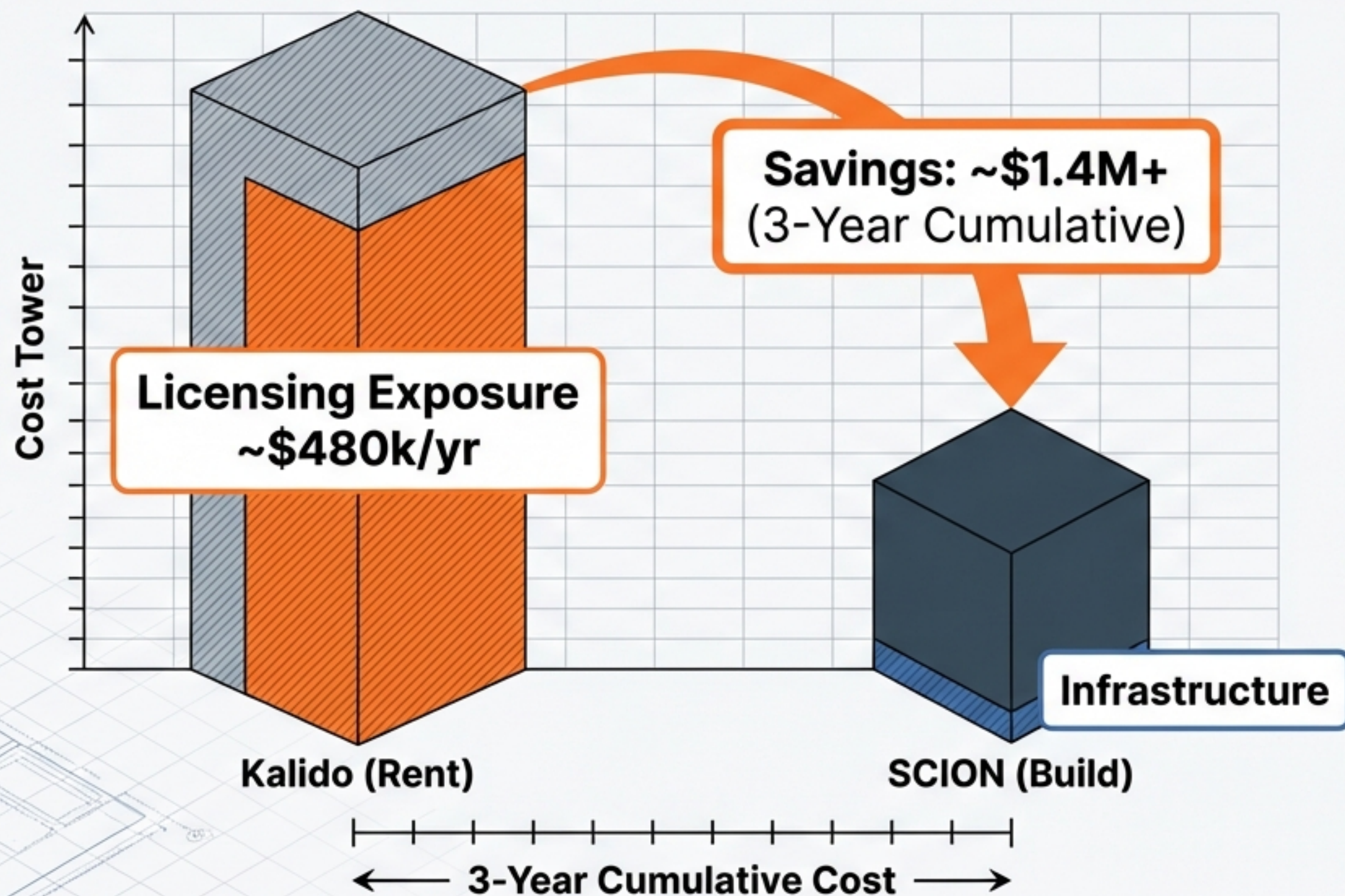
Phase 2 Strategic Potential (Post-DNA Ready)



Modular architecture enables advanced capabilities beyond simple legacy replacement.

teradata NotebookLM

Financial & Strategic Impact



- **Immediate:** Eliminates recurring vendor cost.
- **Asset:** Shifts spend from Renting (OpEx) to Building (IP).
- **Risk:** Eliminates “black-box” dependency.

Recommendation & Next Steps

Financially Justified:
Immediate ROI and
elimination of vendor
costs.

Technically Viable:
MVP validated
Snapshot/Diff logic.

Strategically Essential:
Moves DNA toward
proprietary independence.

**Approve Phase 1 to initiate controlled transition
toward proprietary AI-native governance.**